

RESEARCH ARTICLE

Finite Dimensional Estimation Algebra with State Dimension 3 and Rank 2: Consistency of Wong Matrix

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We study the structure of finite-estimation algebras with state dimension three and rank two arising from smooth nonlinear filtering systems, and establish the constancy of the Wong Ω -matrix in this case. Building on the linear structural results of Shi and Yau (2017), our proof combines Euler-operator techniques, and methods from polynomial dynamical systems. The central idea is to show that any nonzero slope in the affine Wong matrix forces an infinite family of linearly independent elements in the estimation algebra, contradicting finite dimensionality. Consequently, the drift admits a Yau-type representation as the sum of an affine vector field and the gradient of a smooth function. This gives a further structural resolution of Mitter’s conjecture in this setting.¹

Keywords: nonlinear filtering; estimation algebra; Wong matrix; non-maximal rank; Euler operator

17B66; 93E11; 35K15

1. Introduction and main result

Since the development of the Kalman-Bucy filter for linear Gaussian systems in the early 1960s, the search for exact finite-dimensional filters for nonlinear systems has attracted sustained interest. The aim is to describe the conditional distribution of the state exactly through finitely many recursively evolving statistics. The innovations approach and the Duncan-Mortensen-Zakai (DMZ) equation provide general formulations of nonlinear filtering, but they do not in general yield a closed finite-dimensional system (Fujisaki, Kallianpur, and Kunita, 1972; Marcus, 1984). Numerical solution of the resulting equations typically requires approximating the evolving conditional density or its representation (Luo and Yau, 2013). The estimation-algebra approach, introduced by Brockett, Clark, and Mitter (Brockett, 1981; Brockett and Clark, 1980; Mitter, 1979), uses the Lie algebra generated by the filtering operators to investigate when an exact finite-dimensional realization is possible. When this algebra is finite dimensional, the Wei-Norman method can then express the explicit evolution through finitely many ordinary differential equations using Lie algebraic method for time-varying differential equations (Marcus, 1984; Wei and Norman, 1964). The principal advantage of this approach is that it identifies an algebraic mechanism for exact finite-dimensional closure, providing a systematic framework for constructing recursive filters. The study of estimation algebras is therefore of fundamental importance to the theory of finite-dimensional nonlinear filtering.

Following the nonlinear filtering Lie algebra approaches of Beneš and Wong (Beneš, 1981; Wong, 1983), the classification of finite-dimensional estimation algebras of maximal rank was completed by

¹The main theorem was proved by AI Mathematician Liu, Huang, Wang, Li, and Liu (2026), a mathematical proving agent, and its proof was formally verified in Lean, which is publicly available at https://github.com/Pinarinith/AI-Mathematician/tree/main/Mitter_Conjecture/lean. The author takes full responsibility for the correctness of the proof and the entire content of this paper.

Yau and his collaborators (Yau, 2003; Yau and Hu, 2005). For nonmaximal rank, Wu and Yau (Wu and Yau, 2006) classified the case of state dimension two. In state dimension three and linear rank two, Shi and Yau first proved that the Wong matrix is affine (Shi and Yau, 2017) and subsequently established Mitter’s conjecture proposed in ICM 1983, namely that every function element of the estimation algebra is affine (Shi and Yau, 2020). More recent work has substantially extended this picture. Jiao and Yau studied the exact subclass, in which the drift is a gradient and the Wong matrix vanishes, combining structure results in state dimension three and rank two with Riccati and Schrödinger analysis of the associated potentials (Jiao and Yau, 2023). They also extended the affine Wong-matrix theorem to arbitrary state dimension n and rank $n - 1$ (Jiao and Yau, 2024). Under the additional assumption of a constant Wong matrix, Yu, Jiao, and Yau obtained a complete classification for rank $n - 1$ (Yu, Jiao, and Yau, 2024).

The nonmaximal-rank case is more difficult because the algebra contains fewer independent coordinate functions, limiting the control that commutator identities provide over the Wong matrix. Indeed, Jiao and Yau constructed rank- $n - 2$ examples in which the Wong matrix has nonpolynomial smooth elements (Jiao and Yau, 2020). In this paper, we combine techniques of Euler operators and methods from polynomial dynamical systems. We prove that the Wong matrix is constant when the state dimension is three and the linear rank is two, thereby removing the constant-Wong-matrix assumption from the classification of Yu et al. (2024) in this case. The following is our main theorem.

Theorem 1.1 (Main theorem). *Let E be a finite-dimensional estimation algebra of a smooth nonlinear filtering system on $\mathbb{R}^3$, as in (2.2)–(2.3), with state dimension 3 and rank 2. Then the Wong matrix Ω is constant, i.e., $\omega_{12}, \omega_{13}, \omega_{23}$ are constants.*

The paper is organized as follows: Section 2 describes some basic concepts of estimation algebra and well-known results for our proof. In Section 3, we complete the proof within 3 steps:

- (1) We prove that ω_{12} is constant.
- (2) We prove that ω_{13} and ω_{23} do not have x_3 -coordinate.
- (3) We prove that ω_{13} and ω_{23} are constant.

2. Basic concepts and results.

2.1 Basic concepts

The filtering problem we consider in this paper is based on the following model.

$$\begin{aligned} dx(t) &= f(x(t)) dt + g(x(t)) dv(t), & x(0) &= x_0, \\ dy(t) &= h(x(t)) dt + dw(t), & y(0) &= 0. \end{aligned} \tag{2.1}$$

Here x, v, y, w belong to $\mathbb{R}^n, \mathbb{R}^p, \mathbb{R}^m, \mathbb{R}^m$, respectively, where $x(t)$ is the state process and $y(t)$ is the observation process. We assume that f and $h = (q_1, \dots, q_m)$ are C^∞ smooth and $g(x)g(x)^T = I_n$. Let σ denote the normalized conditional density given the observation $\{y(s) : 0 \leq s \leq t\}$, then σ satisfies the well-known DMZ equation

$$d\sigma(t, x) = L_0\sigma(t, x)dt + \sum_{j=1}^m L_j\sigma(t, x)dy_j, \quad \sigma(0, x) = \sigma_0,$$

where $L_j\phi = h_j\phi$ for $1 \leq j \leq m$ and

$$L_0 = \frac{1}{2}\Delta - \sum_{i=1}^n f_i \partial_i - \sum_{i=1}^n \partial_i f_i - \frac{1}{2} \sum_{j=1}^m h_j^2.$$

Equivalently, let

$$D_i = \partial_i - f_i, \quad \eta = \sum_{i=1}^n (\partial_i f_i + f_i^2) + \sum_{j=1}^m h_j^2, \quad L_0 = \frac{1}{2} \sum_{i=1}^n D_i^2 - \frac{1}{2} \eta. \quad (2.2)$$

In the following, we briefly introduce Lie algebra for differential operators and the estimation algebra.

Definition 2.1. For two differential operators X and Y , the Lie bracket $[\cdot, \cdot] : \mathcal{F} \times \mathcal{F} \rightarrow \mathcal{F}$, $[X, Y] := XY - YX$ satisfying the following three axioms:

- (1) The Lie bracket is bilinear;
- (2) $[x, x] = 0$ for all $x \in \mathcal{F}$;
- (3) Jacobi identity: $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$ for all $x, y, z \in \mathcal{F}$.

Definition 2.2. The estimation algebra E of (2.1) is the smallest real Lie algebra containing $L_0, L_1, \dots, L_m$:

$$E = \text{Lie}_{\mathbb{R}} \langle L_0, h_1, \dots, h_m \rangle. \quad (2.3)$$

Its linear rank is $\dim_{\mathbb{R}} L(E)$, where $L(E) = E \cap \text{span}_{\mathbb{R}} \{x_1, x_2, \dots, x_n\}$. A function element of E is a smooth function whose multiplication operator belongs to E .

Definition 2.3. The Wong matrix is $\Omega = (\omega_{ij})_{1 \leq i, j \leq n}$, with

$$\omega_{ij} = \partial_i f_j - \partial_j f_i = [D_j, D_i]. \quad (2.4)$$

In particular, $\omega_{ji} = -\omega_{ij}$.

Definition 2.4. Let $\mathcal{U}_r$ be the vector space of smooth-coefficient differential operators of order at most r with finite index $I_P \subset \mathbb{N}^n$. $P \in \mathcal{U}_r$ can be represented as

$$P = \sum_{(i_1, i_2, \dots, i_n) \in I_P} a_{i_1, i_2, \dots, i_n}(x) \partial_1^{i_1} \partial_2^{i_2} \dots \partial_n^{i_n},$$

where $|i| := \sum_{k=1}^n i_k \leq r$. The order of P is denoted by $\text{ord}(P) = \max_i |i|$.

In this paper, the notation $[D^\alpha]P$ means the coefficient of D^α in operator P . We write $\text{ad}_P Q = [P, Q]$, and $\text{ad}_P^l Q = [P, \text{ad}_P^{l-1} Q]$ for iterated commutators.

2.2 Preliminaries

From now on, we consider the finite-dimensional algebra with state dimension $n = 3$ and linear rank 2. We choose orthonormal coordinates so that there exist c_i such that $x_1 + c_i \in E$, and for any constant c , $x_3 + c \notin E$. We recall the known results and methodologies needed in the proof.

Theorem 2.5 (See Ocone (1981)). *If E is a finite-dimensional estimation algebra and a multiplication operator u belongs to E , then u is a polynomial of degree at most two.*

Lemma 2.6 (See Shi and Yau (2017)). *The elements of the Wong matrix satisfies:*

$$\partial_3\omega_{12} - \partial_2\omega_{13} + \partial_1\omega_{23} = 0. \quad (2.5)$$

For the affine Wong matrix, $\partial_3\omega_{12} = 0$, $\partial_2\omega_{13} = k_2$, and $\partial_1\omega_{23} = h_1$. Consequently $h_1 = k_2$.

Theorem 2.7 (See Shi and Yau (2017)). *Let E be a finite-dimensional estimation algebra. If $P \in E$ has positive differential order r , every coefficient of its ordinary principal symbol $\sigma_r(P)$ is polynomial in the state variables. Equivalently, its highest-order D -normal coefficients are polynomials.*

Theorem 2.8 (See Shi and Yau (2017)). *Suppose that E is finite-dimensional, the state dimension is three, and the rank is two, then*

$$\omega_{12} = b_1x_1 + b_2x_2 + b_0, \quad (2.6)$$

$$\omega_{13} = k_1x_1 + k_2x_2 + k_3x_3 + h_0, \quad (2.7)$$

$$\omega_{23} = h_1x_1 + h_2x_2 + h_3x_3 + k_0, \quad (2.8)$$

where all coefficients are real constants.

Lemma 2.9 (See Shi and Yau (2017)). *Let $\Lambda \in C^\infty(\mathbb{R}^3; \mathbb{R})$. The map $\mathcal{G}_\Lambda(P) = e^\Lambda P e^{-\Lambda}$ is an automorphism of the Lie algebra of differential operators. It fixes multiplication operators and preserves differential order and elements of Wong matrix. In particular, $E_\Lambda := \mathcal{G}_\Lambda(E)$ is isomorphic to E .*

Lemma 2.10. *Let $f \in C^\infty(\mathbb{R}^3; \mathbb{R}^3)$, then there exists $\Lambda \in C^\infty(\mathbb{R}^3; \mathbb{R})$ such that $\tilde{f} = f + \nabla \Lambda$ satisfies*

$$\begin{aligned} \tilde{f}_1 &= 0, \\ \tilde{f}_2 &= \frac{b_1}{2}x_1^2 + b_2x_1x_2 + b_0x_1, \end{aligned} \quad (2.9)$$

$$\tilde{f}_3 = \frac{k_1}{2}x_1^2 + k_2x_1x_2 + k_3x_1x_3 + k_0x_1 + \frac{h_2}{2}x_2^2 + h_3x_2x_3 + h_0x_2. \quad (2.10)$$

Proof. Let $\tilde{f}$ be the polynomial vector field displayed above. Direct differentiation gives $\partial_1\tilde{f}_2 - \partial_2\tilde{f}_1 = \omega_{12}$, $\partial_1\tilde{f}_3 - \partial_3\tilde{f}_1 = \omega_{13}$, $\partial_2\tilde{f}_3 - \partial_3\tilde{f}_2 = \omega_{23}$. Consequently, the one-form $\sum_{i=1}^3(\tilde{f}_i - f_i)dx_i$ is closed. By the Poincaré lemma on $\mathbb{R}^3$, it equals $d\Lambda$ for some globally defined smooth real function Λ . Hence $\tilde{f} = f + \nabla \Lambda$, and $e^\Lambda D_i e^{-\Lambda} = \partial_i - f_i - \partial_i \Lambda = \tilde{D}_i$. $\square$

Lemma 2.11. *Let $g, h \in C^\infty(\mathbb{R}^n)$ and $\alpha, \beta \in \mathbb{N}_0^n$, put $r = |\alpha|$ and $s = |\beta|$. If $r + s \geq 2$, then*

$$[gD^\alpha, hD^\beta] \equiv \sum_{\substack{1 \leq \ell \leq n \\ \alpha_\ell + \beta_\ell > 0}} (\alpha_\ell g \partial_\ell h - \beta_\ell h \partial_\ell g) D^{\alpha+\beta-e_\ell} \pmod{\mathcal{U}_{r+s-2}},$$

where e_ℓ is the ℓ th standard multi-index.

Lemma 2.12. *Let E be an estimation algebra, let $X, Y, Z \in E$, and let $g, h \in C^\infty(\mathbb{R}^n)$. With the notation above, the following identities hold for $1 \leq i, j, k \leq n$:*

$$(1) [XY, Z] = X[Y, Z] + [X, Z]Y.$$

$$(2) [L_0, g] = \sum_{a=1}^n \frac{\partial g}{\partial x_a} D_a + \frac{1}{2} \Delta g.$$

$$(3) [L_0, D_i] = \sum_{a=1}^n \omega_{ia} D_a + \frac{1}{2} \sum_{a=1}^n \frac{\partial \omega_{ia}}{\partial x_a} + \frac{1}{2} \frac{\partial \eta}{\partial x_i}.$$

$$\begin{aligned}
(4) \quad [gD_i, hD_j] &= gh\omega_{ji} + g\frac{\partial h}{\partial x_i}D_j - h\frac{\partial g}{\partial x_j}D_i, \text{ where } \omega_{ji} = [D_i, D_j]. \\
(5) \quad [gD_i^2, h] &= 2g\frac{\partial h}{\partial x_i}D_i + g\frac{\partial^2 h}{\partial x_i^2}. \\
(6) \quad [D_i^2, hD_j] &= 2\frac{\partial h}{\partial x_i}D_iD_j + 2h\omega_{ji}D_i + \frac{\partial^2 h}{\partial x_i^2}D_j + h\frac{\partial \omega_{ji}}{\partial x_i}. \\
(7) \quad [D_i^2, D_j^2] &= 4\omega_{ji}D_jD_i + 2\frac{\partial \omega_{ji}}{\partial x_j}D_i + 2\frac{\partial \omega_{ji}}{\partial x_i}D_j + \frac{\partial^2 \omega_{ji}}{\partial x_i \partial x_j} + 2\omega_{ji}^2. \\
(8) \quad [D_k^2, hD_iD_j] &= 2\frac{\partial h}{\partial x_k}D_kD_iD_j + 2h\omega_{jk}D_iD_k + 2h\omega_{ik}D_kD_j + \frac{\partial^2 h}{\partial x_k^2}D_iD_j \\
&\quad + 2h\frac{\partial \omega_{jk}}{\partial x_i}D_k + h\frac{\partial \omega_{jk}}{\partial x_k}D_i + h\frac{\partial \omega_{ik}}{\partial x_k}D_j + h\frac{\partial^2 \omega_{jk}}{\partial x_i \partial x_k}. \\
(9) \quad [gD_iD_j, hD_k] &= g\frac{\partial h}{\partial x_j}D_iD_k + g\frac{\partial h}{\partial x_i}D_jD_k - h\frac{\partial g}{\partial x_k}D_iD_j \\
&\quad + gh\omega_{kj}D_i + gh\omega_{ki}D_j + g\frac{\partial^2 h}{\partial x_i \partial x_j}D_k + gh\frac{\partial \omega_{kj}}{\partial x_i}.
\end{aligned}$$

3. Proof of the main theorem

3.1 ω_{12} is constant

First, we consider

$$H_0 = [L_0, D_1] - \frac{1}{2}(b_2 + k_3) = \omega_{12}D_2 + \omega_{13}D_3 + \frac{1}{2}\eta_1 \in E, \quad (3.1)$$

$$H_1 = [L_0, D_2] - \frac{1}{2}(-b_1 + h_3) = -\omega_{12}D_1 + \omega_{23}D_3 + \frac{1}{2}\eta_2 \in E. \quad (3.2)$$

For $B \in E \cap \mathcal{U}_2$, write

$$B = a_{11}D_1^2 + a_{12}D_1D_2 + a_{22}D_2^2 + \cdots \pmod{\mathcal{U}_1},$$

where the omitted second-order terms contain D_3 . Direct computation gives $[[B, x_1], x_1] = 2a_{11}$, $[[B, x_1], x_2] = a_{12}$, $[[B, x_2], x_2] = 2a_{22}$. These functions belong to E . By the preceding result on function elements, $a_{11}, a_{12}, a_{22} \in \text{span}_{\mathbb{R}}\{1, x_1, x_2\}$.

Lemma 3.1. *Let $C \in E \cap \mathcal{U}_2$. If, for $i = 1$ or $i = 2$,*

$$C = (Ax_i + c)D_i^2 + \cdots \pmod{\mathcal{U}_1},$$

where $A, c \in \mathbb{R}$, then $A = 0$.

Proof. It suffices to consider $i = 1$. Define $P_0 = [L_0, C]$, $P_{r+1} = [P_r, C]$ ($r \geq 0$). Each P_r belongs to E . The commutator formulas give

$$P_0 = AD_1^3 + \cdots \pmod{\mathcal{U}_2}$$

and, inductively,

$$P_r = \frac{(r+2)!}{2} A^{r+1} D_1^{r+3} + \cdots \pmod{\mathcal{U}_{r+2}}.$$

Here the terms containing D_2 or D_3 with order $r+3$ are omitted. Indeed, since $[D_1^{r+3}, C] = (r+3)AD_1^{r+4} + \cdots \pmod{\mathcal{U}_{r+3}}$, commuting two omitted terms leaves at least one D_2 or D_3 factor at order $r+4$. Commuting an omitted term with a displayed term can remove its last such factor only by differentiating $Ax_1 + c$, or the constant displayed coefficient of P_r , in x_2 or x_3 ; these derivatives vanish. The terms arising from $[D_i, D_j] = \omega_{ji}$ have lower differential order and are included in $\mathcal{U}_{r+3}$.

If $A \neq 0$, P_r has order $r+3$ lies in E , contradicting finite-dimensionality of E . The case $i = 2$ is identical after interchanging x_1 and x_2 . $\square$

Lemma 3.2. *Under the assumptions of the main theorem, suppose that $b_1 = 0$ and $b_2 \neq 0$, so that*

$$\omega_{12} = b_2 x_2 + b_0.$$

Then, for every $B \in E \cap \mathcal{U}_2$, the coefficients of $D_1 D_2$ and D_2^2 are constant.

Proof. Direct calculation gives

$$\begin{aligned} [H_0, B] &= \omega_{12}(\partial_2 a_{11})D_1^2 + (\omega_{12}\partial_2 a_{12} - b_2 a_{12})D_1 D_2 + (\omega_{12}\partial_2 a_{22} - 2b_2 a_{22})D_2^2 + \cdots \pmod{\mathcal{U}_1}, \\ [H_1, B] &= (-\omega_{12}\partial_1 a_{11} + b_2 a_{12})D_1^2 + (-\omega_{12}\partial_1 a_{12} + 2b_2 a_{22})D_1 D_2 - \omega_{12}(\partial_1 a_{22})D_2^2 + \cdots \pmod{\mathcal{U}_1}. \end{aligned}$$

The omitted second-order terms again contain D_3 . Terms involving D_3 in B do not contribute to the three displayed coefficients, because $\partial_3 \omega_{12} = \partial_3 a_{11} = \partial_3 a_{12} = \partial_3 a_{22} = 0$. Both commutators belong to $E \cap \mathcal{U}_2$.

Since a_{11} is affine,

$$[H_0, B] - b_2 B = (-b_2(\partial_1 a_{11})x_1 + b_0\partial_2 a_{11} - b_2 a_{11}(0))D_1^2 + \cdots \pmod{\mathcal{U}_1},$$

where the omitted terms are the other second-order terms. Lemma 3.1 gives $\partial_1 a_{11} = 0$. This holds for every $B \in E \cap \mathcal{U}_2$.

Similarly, $[H_1, B] = -(\partial_1 a_{22})(b_2 x_2 + b_0)D_2^2 + \cdots \pmod{\mathcal{U}_1}$, where the omitted terms are the other second-order terms. The same lemma gives $\partial_1 a_{22} = 0$. Applying it once more to B gives $\partial_2 a_{22} = 0$. Thus a_{22} is constant.

The coefficient of D_1^2 in $[H_1, B]$ is now $b_2 a_{12}$. By the conclusion already obtained for every element of $E \cap \mathcal{U}_2$, this coefficient is independent of x_1 . Hence $\partial_1 a_{12} = 0$. It remains to prove $\partial_2 a_{12} = 0$.

Define $P_0 = [L_0, B]$, $P_{r+1} = [P_r, B]$ ($r \geq 0$). Since a_{11}, a_{12} are affine functions of x_2 and a_{22} is constant, direct calculation gives

$$P_0 = (\partial_2 a_{11})D_1^2 D_2 + (\partial_2 a_{12})D_1 D_2^2 + \cdots \pmod{\mathcal{U}_2}.$$

Repeated application of the same commutator formulas yields, for $r \geq 1$,

$$\begin{aligned} P_r &= 2^r (\partial_2 a_{12})^{r+1} D_1^{r+1} D_2^2 \\ &\quad + (2^{r+1} - 1)(\partial_2 a_{11})(\partial_2 a_{12})^r D_1^{r+2} D_2 \\ &\quad + (2^r - 1)(\partial_2 a_{11})^2 (\partial_2 a_{12})^{r-1} D_1^{r+3} + \cdots \pmod{\mathcal{U}_{r+2}}. \end{aligned}$$

To check the induction, commute each of the three displayed monomials with B . Only differentiation of a_{11} and a_{12} with respect to x_2 contributes to terms of the next order that do not contain D_3 . The omitted terms cannot contribute such terms, because all displayed coefficients are independent of x_3 .

If $\partial_2 a_{12} \neq 0$, the coefficient of $D_1^{r+1} D_2^2$ is nonzero for every r , so the orders of the elements $P_r \in E$ are unbounded. This contradicts finite-dimensionality. a_{12} is constant as well. $\square$

Theorem 3.3. *Under the assumptions of the main theorem, $b_1 = b_2 = 0$, so $\omega_{12} = b_0$.*

Proof. Suppose that $(b_1, b_2) \neq (0, 0)$. Make the orthogonal change of coordinates

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{1}{\sqrt{b_1^2 + b_2^2}} \begin{pmatrix} b_2 & b_1 \\ -b_1 & b_2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad x_3 = y_3.$$

This transformation preserves the second-order part of L_0 and transforms ω_{12} into $\sqrt{b_1^2 + b_2^2} y_2 + b_0$. The transformed algebra is isomorphic to E , and ω_{13}, ω_{23} stay affine. After renaming the coordinates and coefficients, we may therefore assume $b_1 = 0, b_2 > 0, \omega_{12} = b_2 x_2 + b_0$. Direct calculation gives

$$[D_1, H_0] = k_1 D_3 \pmod{\mathcal{U}_0}, \quad [D_1, H_1] = k_2 D_3 \pmod{\mathcal{U}_0}.$$

Since $[D_3, x_1] = [D_3, x_2] = 0$ and $\partial_3 \omega_{12} = 0$, it follows that

$$\begin{aligned} [[D_1, H_0], x_i] &= [[D_1, H_1], x_i] = 0 \quad (i = 1, 2), \\ [\omega_{12}, [D_1, H_1]] &= 0. \end{aligned} \tag{3.3}$$

The constant terms subtracted in (3.1)–(3.2) do not affect the following commutators. By the Jacobi identity,

$$[D_2, H_0] - [D_1, H_1] = [L_0, \omega_{12}] = b_2 D_2.$$

Using $[D_2, D_1] = \omega_{12}$ and $[H_0, \omega_{12}] = b_2 \omega_{12}$, we obtain

$$[D_2, [D_1, H_0]] - [D_1, [D_1, H_1]] = [D_1, b_2 D_2] + [\omega_{12}, H_0] = -2b_2 \omega_{12}. \tag{3.4}$$

Next, direct calculation gives

$$[L_0, H_0] = b_2 D_2^2 + k_1 D_1 D_3 + k_2 D_2 D_3 + k_3 D_3^2 \pmod{\mathcal{U}_1}.$$

All displayed coefficients are constant. Since $[L_0, D_i D_j] = [L_0, D_i] D_j + D_i [L_0, D_j] \in \mathcal{U}_2$, and the commutator of L_0 with a first-order operator has order at most two, we have $[L_0, [L_0, H_0]] \in E \cap \mathcal{U}_2$.

Using $[H_0, x_1] = 0, [H_0, x_2] = \omega_{12}$, and the Jacobi identity, we obtain

$$\begin{aligned} [[L_0, H_0], x_1] &= [D_1, H_0], \\ [[L_0, H_0], x_2] &= [D_1, H_1] + 2b_2 D_2, \\ [[[L_0, [L_0, H_0]], x_1], x_2] &= 2[D_2, [D_1, H_0]], \\ [[[L_0, [L_0, H_0]], x_2], x_2] &= 2[D_2, [D_1, H_1]]. \end{aligned}$$

The left-hand sides of the last two equations are, respectively, the coefficient of $D_1 D_2$ and twice the coefficient of D_2^2 in $[L_0, [L_0, H_0]] \in E \cap \mathcal{U}_2$. By Lemma 3.2, these coefficients are constant.

Consequently,

$$[D_2, [D_1, H_0]] \quad \text{and} \quad [D_2, [D_1, H_1]]$$

are constant multiplication operators.

Finally, take the commutator of both sides of (3.4) with D_2 :

$$[D_2, [D_1, [D_1, H_1]]] = [D_1, [D_2, [D_1, H_1]]] + [\omega_{12}, [D_1, H_1]] = [D_1, [D_2, [D_1, H_1]]],$$

where the last equality follows from (3.3). The constant multiplication operators above commute with D_1 and D_2 . Therefore,

$$0 = [D_2, [D_2, [D_1, H_0]]] - [D_1, [D_2, [D_1, H_1]]] = [D_2, -2b_2\omega_{12}] = -2b_2^2 1.$$

This contradicts $b_2 > 0$ and proves the theorem. $\square$

3.2 $h_3 = k_3 = 0$.

By Theorem 3.3, $b_1 = b_2 = 0$. Suppose that $\mu = k_3^2 + h_3^2 > 0$. By Lemmas 2.10 and 2.9, we may use the drift in (2.9)–(2.10), retaining the notation E, D_i, L_0, H_i . In particular, $f_1 = 0$ and $f_2 = b_0 x_1$.

Choose the affine coordinate

$$t = \frac{k_3\omega_{13} + h_3\omega_{23}}{\mu} = x_3 + \alpha x_1 + \beta x_2 + \delta,$$

where $\alpha = \frac{k_3 k_1 + h_3 k_2}{\mu}$, $\beta = \frac{k_3 k_2 + h_3 h_2}{\mu}$, $\delta = \frac{k_3 k_0 + h_3 h_0}{\mu}$. The coordinate derivatives in (x_1, x_2, t) are $\nabla_1 = \partial_1 - \alpha \partial_3$, $\nabla_2 = \partial_2 - \beta \partial_3$, $\partial_t = \partial_3$, where ∂_i^μ denotes differentiation in the original coordinates. Then, the Euler operator

$$J := \frac{k_3(H_0 - b_0 D_2) + h_3(H_1 + b_0 D_1)}{\mu} = t \partial_3 + r(x_1, x_2, t) \in E$$

for a smooth function r . Take

$$\Lambda(x_1, x_2, t) = \int_0^t \frac{r(x_1, x_2, s) - r(x_1, x_2, 0)}{s} ds,$$

where the integrand extends smoothly across $s = 0$. Writing $\hat{P} = e^\Lambda P e^{-\Lambda}$ and $\hat{E} = e^\Lambda E e^{-\Lambda}$, we obtain $\hat{J} = t \partial_3 + r(x_1, x_2, 0)$, and $\hat{D}_1 = \nabla_1 + \alpha \partial_3 - \partial_1 \Lambda$, $\hat{D}_2 = \nabla_2 + \beta \partial_3 - \partial_2 \Lambda$.

To extract the required terms involving ∂_3 as elements of the estimation algebra, we establish the following lemma. It shows that projection onto each weight component preserves membership in the algebra for the operators in $\hat{E}$.

Lemma 3.4. *Let $\hat{E}, \hat{D}_1, \hat{D}_2, \hat{L}_0$ be defined as above, and suppose that*

$$\hat{J} = t \partial_t + r_0(x_1, x_2) \in \hat{E}.$$

For each $P \in \{\hat{D}_1, \hat{D}_2, \hat{L}_0\}$, write

$$P = \sum_{\substack{i, j, q \geq 0 \\ i+j+q \leq 2}} a_{ijq}(x_1, x_2, t) \partial_{x_1}^i \partial_{x_2}^j \partial_t^q.$$

(1) Then every coefficient is polynomial in t :

$$a_{ijq}(x_1, x_2, t) = \sum_{p \geq 0} a_{ijqp}(x_1, x_2) t^p, \quad a_{ijqp} \in C^\infty(\mathbb{R}^2).$$

(2) Moreover, define the **weight** of a nonzero term by $\text{wt}(a_{ijqp}(x_1, x_2) t^p \partial_{x_1}^i \partial_{x_2}^j \partial_t^q) = p - q$. Let P_w denote the sum of all terms of weight w in P , then for every $w \in \mathbb{Z}$, there exists $\pi_w \in \mathbb{R}[s]$ such that

$$P_w = \pi_w(\text{ad}_{\hat{f}})P \in \hat{E}.$$

Proof. Direct calculation gives

$$[t\partial_t, a\partial_{x_1}^i \partial_{x_2}^j \partial_t^q] = (t\partial_t a - qa)\partial_{x_1}^i \partial_{x_2}^j \partial_t^q.$$

In a commutator with r_0 , at least one derivative in x_1, x_2 falls on r_0 . Thus commutation with r_0 lowers the differential order in these variables. Since P has order at most two, $(\text{ad}_{r_0})^3 P = 0$. Moreover, r_0 is independent of t , so the Jacobi identity gives $[\text{ad}_{t\partial_t}, \text{ad}_{r_0}] = \text{ad}_{[t\partial_t, r_0]} = 0$.

Let χ be the characteristic polynomial of $\text{ad}_{\hat{f}}$ on $\hat{E}$. By the Cayley–Hamilton theorem, $\chi(\text{ad}_{\hat{f}})P = 0$. Expanding this polynomial in the two commuting adjoint maps and using $(\text{ad}_{r_0})^3 P = 0$, we obtain

$$0 = \chi(\text{ad}_{t\partial_t})P + \chi'(\text{ad}_{t\partial_t})[r_0, P] + \frac{1}{2}\chi''(\text{ad}_{t\partial_t})[r_0, [r_0, P]].$$

Because the adjoint maps commute, this identity can be substituted repeatedly. Each substitution introduces at least one further commutator with r_0 . After three substitutions, every resulting term vanishes. Hence $[\chi(\text{ad}_{t\partial_t})]^3 P = 0$.

The first displayed identity shows that $\text{ad}_{t\partial_t}$ does not mix different derivative monomials. Comparing their coefficients gives

$$[\chi(t\partial_t - q)]^3 a_{ijq} = 0.$$

Applying Lemma A.1 for each fixed (x_1, x_2) yields

$$a_{ijq}(x_1, x_2, t) = \sum_{\substack{p \geq 0 \\ \chi(p-q)=0}} \frac{\partial_t^p a_{ijq}(x_1, x_2, 0)}{p!} t^p.$$

There are only finitely many such p , independently of (x_1, x_2) , and the displayed coefficients are smooth. This proves (1).

For (2), write $P = \sum_w P_w$, a finite sum. By the definition of weight,

$$[t\partial_t, P_w] = wP_w.$$

Commutation with r_0 changes neither the power of t nor the number of ∂_t factors, so it preserves weight. It also lowers the differential order in x_1, x_2 . Consequently,

$$(\text{ad}_{\hat{f}} - w)^3 P_w = (\text{ad}_{r_0})^3 P_w = 0.$$

Fix an occurring weight w . Since the polynomials $(s - v)^3$ for distinct occurring weights v are

pairwise coprime, the Chinese remainder theorem gives $\pi_w \in \mathbb{R}[s]$ satisfying

$$\begin{aligned}\pi_w(s) &\equiv 1 \pmod{(s-w)^3}, \\ \pi_w(s) &\equiv 0 \pmod{(s-v)^3} \quad \text{for every other occurring weight } v.\end{aligned}$$

The preceding annihilation identities therefore imply

$$\pi_w(\text{ad}_{\hat{J}})P = \sum_v \pi_w(\text{ad}_{\hat{J}})P_v = P_w.$$

Since $\hat{J} \in \hat{E}$, its adjoint action preserves $\hat{E}$, and hence $P_w \in \hat{E}$. For a weight that does not occur, take $\pi_w = 0$. $\square$

Theorem 3.5. *Under the assumptions of the main theorem, $k_3 = h_3 = 0$.*

Proof. By Lemma 3.4, $\partial_1\Lambda$ and $\partial_2\Lambda$ are polynomial in t , and their positive-weight components are multiplication operators in $\hat{E}$. Since the coordinate change leaves x_1, x_2 fixed and gauge conjugation fixes multiplication operators, we also have $\hat{E} \cap C^\infty(\mathbb{R}^3) = \text{span}_{\mathbb{R}}\{1, x_1, x_2\}$. Hence $\partial_1\Lambda$ and $\partial_2\Lambda$ are independent of t , while $\partial_3\Lambda = \rho(x_3) = \rho(t - \alpha x_1 - \beta x_2 - \delta)$ for some $\rho \in C^\infty(\mathbb{R})$.

Using (2.10), direct expansion gives

$$\begin{aligned}\hat{L}_0 &= \underbrace{\frac{1}{2}(\nabla_1^2 + \nabla_2^2)}_{\text{weight } 0} + \underbrace{\alpha\nabla_1\partial_3 + \beta\nabla_2\partial_3}_{\text{weight } -1} + \underbrace{\frac{1 + \alpha^2 + \beta^2}{2}\partial_3^2}_{\text{weight } -2} \\ &\quad + \underbrace{a_1\nabla_1 + a_2\nabla_2 - ((k_3x_1 + h_3x_2)t + \rho(t - \alpha x_1 - \beta x_2 - \delta) + b)\partial_3 + u(x_1, x_2, t)}_{\text{weight } \geq 0, \text{ depending on the order of } \rho}.\end{aligned}\tag{3.5}$$

where a_1, a_2, b depend only on x_1, x_2 . Lemma 3.4 gives $u \in C^\infty(\mathbb{R}^2)[t]$, $\rho \in \mathbb{R}[t]$ is a polynomial, and the -2 weight term is

$$(\hat{L}_0)_{-2} = \frac{1 + \alpha^2 + \beta^2}{2}\partial_3^2 \in \hat{E}, \quad \partial_3^2 \in \hat{E}.$$

Case 1: $\deg \rho = d \geq 2$. Write $\rho(s) = a_d s^d + \dots$, with $a_d \neq 0$. Then

$$(\hat{L}_0)_{d-1} = -a_d t^d \partial_3 \pmod{\mathcal{U}_0}$$

belongs to $\hat{E}$. Start with $P_2 = \partial_3^2$. If $P_m = \partial_3^m \pmod{\mathcal{U}_{m-1}}$ belongs to $\hat{E}$, the commutator formulas give

$$\begin{aligned}[(\hat{L}_0)_{d-1}, P_m] &= m a_d t^{d-1} \partial_3^m \pmod{\mathcal{U}_{m-1}}, \\ [P_m, [(\hat{L}_0)_{d-1}, P_m]] &= m^2 a_d d(d-1) t^{d-2} \partial_3^{2m-1} \pmod{\mathcal{U}_{2m-2}}.\end{aligned}$$

Commuting $d-2$ times with ∂_3^2 gives

$$\text{ad}_{\partial_3^2}^{d-2}([P_m, [(\hat{L}_0)_{d-1}, P_m]]) = 2^{d-2} m^2 a_d d! \partial_3^{2m+d-3} \pmod{\mathcal{U}_{2m+d-4}}.$$

Dividing by the displayed nonzero constant produces P_{2m+d-3} with the same required form. Since $2m + d - 3 > m$, iteration gives elements of unbounded differential order, contradicting finite-dimensionality.

Case 2: $\rho(s) = cs + c_0$. This includes $\rho = 0$. The weight-zero component is

$$(\widehat{L}_0)_0 = \frac{1}{2}(\nabla_1^2 + \nabla_2^2) + a_1\nabla_1 + a_2\nabla_2 - (k_3x_1 + h_3x_2 + c)t\partial_3 + u_0(x_1, x_2) \in \widehat{E}.$$

If $\partial_3^m \in \widehat{E}$ for $m \geq 2$, then

$$\frac{[(\widehat{L}_0)_0, \partial_3^m] - mc\partial_3^m}{m} = (k_3x_1 + h_3x_2)\partial_3^m \in \widehat{E}.$$

A further calculation gives

$$[(k_3x_1 + h_3x_2)\partial_3^m, [(\widehat{L}_0)_0, (k_3x_1 + h_3x_2)\partial_3^m]] = -(k_3^2 + h_3^2)\partial_3^{2m} = -\mu\partial_3^{2m} \in \widehat{E}.$$

which is a contradiction against finite-dimensionality. Therefore $k_3 = h_3 = 0$. $\square$

4. $k_1 = k_2 = h_1 = h_2 = 0$

Lemma 4.1. *Let $p, q \in \mathbb{R}[\tau]$. If $pq' - p'q$ is a nonzero constant, then*

$$\text{span}_{\mathbb{R}}\{p, q\} = \text{span}_{\mathbb{R}}\{1, \tau\}.$$

Proof. The nonzero Wronskian implies that p, q are linearly independent. If their degrees coincide, subtract a constant multiple of one from the other to cancel its leading term. After interchanging them if necessary, their degrees satisfy $r < s$. Their Wronskian has degree $r + s - 1$, with nonzero leading coefficient $(s - r)\text{lc}(p)\text{lc}(q)$, where $\text{lc}(p)$ means the leading coefficient of polynomial p . Thus $r + s = 1$, so $r = 0$ and $s = 1$, proving the assertion. $\square$

Theorem 4.2. *Under the assumptions of the main theorem, $k_1 = k_2 = h_2 = 0$.*

Sketch of proof. We argue by contradiction. If the remaining coefficient matrix $S_n = (\text{ad}_{\widetilde{L}_0})^n(\partial_1) \in \widetilde{E}$ is nonzero, an orthogonal coordinate change and a translation reduce the problem to $k_1 \neq 0$, $k_2 = k_0 = 0$. We can restrict the x_3 -dependence of η and produces a first-order element of the form $\partial_3 + g(x_1, x_2) + \varepsilon(x_3)$.

A gauge conjugation removes ε (which is a polynomial by Theorem 2.7) from this first-order element while preserving finite-dimensionality. Further commutators, together with the function-element theorem, show that the remaining scalar coefficient is also polynomial. Consequently, every coefficient of $\widetilde{L}_0$ is polynomial in x_3 .

We then consider a sequence of Lie algebra elements

$$S_n = (\text{ad}_{\widetilde{L}_0})^n(\partial_1) \in \widetilde{E}.$$

Their coefficients remain polynomial in x_3 , so each S_n has finite "weighted-degrees". Since their span is finite-dimensional, these weights must have a common upper bound.

We contradict this bound in two cases. If $\deg \varepsilon \geq 3$, direct commutator calculations give terms whose nonzero highest weighted-degrees increases without bound. If $\deg \varepsilon \leq 2$, these terms satisfy a polynomial recurrence described by an associated dynamical system. If the recurrence vanished at some stage, suitable trajectories would yield two linearly independent polynomial solutions of

$$u'' = k_1\zeta(\tau)u, \quad \zeta(0) = 1.$$

Their Wronskian is a nonzero constant, so Lemma 4.1 would force their span to contain a nonzero constant solution, contradicting $k_1 \neq 0$. Thus the highest-weight terms never vanish, and the weighted-degrees again grow without bound, deriving contradictions.

Proof. By Theorems 3.3 and 3.5, $\omega_{12} = b_0$ and $k_3 = h_3 = 0$. By Lemma 2.6 we have $h_1 = k_2$, suppose that the symmetry matrix

$$\begin{pmatrix} k_1 & k_2 \\ k_2 & h_2 \end{pmatrix} \neq 0.$$

An orthogonal change of (x_1, x_2) diagonalizes this matrix. Choose a nonzero eigenvalue as the first diagonal entry and translate x_1 to remove the constant term in ω_{13} . After renaming the coordinates and coefficients, we have

$$k_1 \neq 0, \quad k_2 = k_0 = 0, \quad \omega_{13} = k_1 x_1, \quad \omega_{23} = h_2 x_2 + h_0.$$

By Lemmas 2.10 and 2.9, we may also assume $f_1 = 0, f_2 = b_0 x_1, f_3 = \frac{k_1}{2} x_1^2 + \frac{h_2}{2} x_2^2 + h_0 x_2$, retaining the notation E, D_i, L_0, H_i, η .

Step 1: Determine the x_3 -dependence of η .

Direct calculation gives

$$[D_1, H_0] = k_1 D_3 - b_0^2 - k_1^2 x_1^2 + \frac{1}{2} \eta_{11}, \quad (4.1)$$

$$[D_2, H_0] = -k_1 x_1 (h_2 x_2 + h_0) + \frac{1}{2} \eta_{12}, \quad (4.2)$$

$$[D_2, H_1] = h_2 D_3 - b_0^2 - (h_2 x_2 + h_0)^2 + \frac{1}{2} \eta_{22}, \quad (4.3)$$

and hence

$$[D_1, [D_1, H_0]] = \frac{1}{2} \partial_1^3 (\eta - f_3^2), \quad (4.4)$$

$$[D_2, [D_1, H_0]] = \frac{1}{2} \partial_1^2 \partial_2 (\eta - f_3^2), \quad (4.5)$$

$$[D_2, [D_2, H_0]] = \frac{1}{2} \partial_1 \partial_2^2 (\eta - f_3^2), \quad (4.6)$$

$$[D_2, [D_2, H_1]] = \frac{1}{2} \partial_2^3 (\eta - f_3^2). \quad (4.7)$$

These are function elements of E , and therefore belong to $\text{span}_{\mathbb{R}}\{1, x_1, x_2\}$.

Set $\eta_0(x_1, x_2) = \eta(x_1, x_2, 0)$. Since f_3 is independent of x_3 , the preceding identities 4.4-4.7 show that all third-order derivatives of $\eta - \eta_0$ with respect to x_1, x_2 vanish. Taylor's formula therefore gives

$$\begin{aligned} \eta &= \eta_0(x_1, x_2) + a_{11}(x_3)x_1^2 + a_{12}(x_3)x_1x_2 + a_{22}(x_3)x_2^2 \\ &\quad + \ell_1(x_3)x_1 + \ell_2(x_3)x_2 + e(x_3), \end{aligned}$$

where all six coefficient functions are smooth and vanish at $x_3 = 0$.

Moreover, $[D_2, H_0]$ and $h_2[D_1, H_0] - k_1[D_2, H_1]$ are function elements of E and hence are independent of x_3 . The equations 4.1-4.3 consequently imply that η_{12} and $h_2\eta_{11} - k_1\eta_{22}$ are independent of x_3 . Substituting the expansion above gives

$$a'_{12}(x_3) = 0, \quad h_2 a'_{11}(x_3) - k_1 a'_{22}(x_3) = 0.$$

Since $a_{11}(0) = a_{12}(0) = a_{22}(0) = 0$, it follows that $a_{12} = 0, h_2 a_{11} = k_1 a_{22}$. As $k_1 \neq 0$, define $\varepsilon(x_3) = \frac{a_{11}(x_3)}{k_1}$. Then $a_{11} = k_1 \varepsilon$ and $a_{22} = h_2 \varepsilon$. Using $f_3 = \frac{k_1}{2} x_1^2 + \frac{h_2}{2} x_2^2 + h_0 x_2$, we obtain $a_{11} x_1^2 +$

$a_{22}x_2^2 = 2\varepsilon f_3 - 2h_0\varepsilon x_2$. Replacing ℓ_2 by $\ell_2 - 2h_0\varepsilon$ and retaining the same notation, we thus have

$$\eta = \eta_0(x_1, x_2) + 2\varepsilon(x_3)f_3 + \ell_1(x_3)x_1 + \ell_2(x_3)x_2 + e(x_3).$$

Here $\varepsilon, \ell_1, \ell_2, e$ are smooth and still vanish at $x_3 = 0$.

The identity for $[D_1, H_0]$ now gives

$$T = \frac{1}{k_1}[D_1, H_0] = \partial_3 + g(x_1, x_2) + \varepsilon(x_3) \in E$$

for some $g \in C^\infty(\mathbb{R}^2)$. Direct calculation yields

$$[T, H_0 - b_0 D_2] = \frac{1}{2}\ell'_1(x_3), \quad [T, H_1 + b_0 D_1] = \frac{1}{2}\ell'_2(x_3).$$

Both are function elements depending only on x_3 , so ℓ'_1 and ℓ'_2 are constants. Since $\ell_1(0) = \ell_2(0) = 0$, we obtain $\ell_1(x_3) = c_1 x_3$ and $\ell_2(x_3) = c_2 x_3$. Consequently,

$$\eta = \eta_0(x_1, x_2) + 2\varepsilon(x_3)f_3 + e(x_3) + x_3(c_1 x_1 + c_2 x_2), \quad c_1, c_2 \in \mathbb{R}. \quad (4.8)$$

Step 2: Obtain polynomial coefficients in x_3 .

Choose $\Lambda = \Lambda(x_3)$ with $\Lambda' = \varepsilon$, and write $\tilde{P} = e^\Lambda P e^{-\Lambda}$ and $\tilde{E} = e^\Lambda E e^{-\Lambda}$. Then $\partial_1 \in \tilde{E}$ and $\tilde{T} = \partial_3 + g(x_1, x_2) \in \tilde{E}$. Set $r = e - \varepsilon^2$. Expanding $\tilde{L}_0$ using (4.8) gives

$$\begin{aligned} \tilde{L}_0 &= \frac{1}{2}(\partial_1^2 + \partial_2^2 + \partial_3^2) - b_0 x_1 \partial_2 - (f_3 + \varepsilon) \partial_3 \\ &\quad + V_0(x_1, x_2) + x_3 V_1(x_1, x_2) - \frac{1}{2}(\varepsilon'(x_3) + r(x_3)). \end{aligned} \quad (4.9)$$

Furthermore,

$$[\tilde{L}_0, \tilde{T}] = (\partial_1 g) \partial_1 + (\partial_2 g) \partial_2 + \varepsilon'(x_3) \partial_3 \pmod{\mathcal{U}_0}.$$

Transporting this operator back to E and applying Theorem 2.7 shows that ε' is polynomial; if the operator has order zero, the displayed coefficients vanish. Hence $\varepsilon \in \mathbb{R}[x_3]$.

Let $d = \deg \varepsilon$, with $d = 0$ if $\varepsilon = 0$, and let $m = \max\{2, d\}$. The commutator formulas give

$$[g, [g, \tilde{L}_0]] = (\partial_1 g)^2 + (\partial_2 g)^2, \quad [\partial_3, [g, \tilde{L}_0]] = 0.$$

Since $[\partial_3, g] = 0$, it follows that

$$\text{ad}_T^m(\tilde{L}_0) + \varepsilon^{(m)} \tilde{T} = \varepsilon^{(m)} g + \mathbf{1}_{\{m=2\}}((\partial_1 g)^2 + (\partial_2 g)^2) - \frac{1}{2}r^{(m)}(x_3).$$

Here $\varepsilon^{(m)}$ is constant, so the left-hand side belongs to $\tilde{E}$. It is a multiplication operator, and therefore is independent of x_3 . Thus $r^{(m)}$ is constant and

$$r \in \mathbb{R}[x_3], \quad \deg r \leq \max\{2, d\} \quad (r \neq 0).$$

Step 3: Exclude $d \geq 3$ by increasing weighted-degrees.

For the remaining steps, consider finite-order differential operators whose coefficients are smooth in x_1, x_2 and polynomial in x_3 . Define the weighted degree of a nonzero term by

$$\nu(a(x_1, x_2)x_3^p\partial_1^{q_1}\partial_2^{q_2}\partial_3^{q_3}) = p + q_1 + q_2 + 2q_3,$$

where $a \in C^\infty(\mathbb{R}^2)$ and $p, q_1, q_2, q_3 \in \mathbb{N}_0$. For a nonzero operator P , let $\nu(P)$ be the largest weighted degree of its nonzero terms in normal form, and set $\nu(0) = -\infty$. The product rule gives, for operators P polynomial in x_3 and integers $s \geq 0$,

$$\begin{aligned}\nu([\tfrac{1}{2}\Delta - f_3\partial_3, P]) &\leq \nu(P) + 1, \\ \nu([x_3^s\partial_3, P]) &\leq \nu(P) + s - 1, \\ \nu([x_3^s, P]) &\leq \nu(P) + s - 3, \\ \nu([-b_0x_1\partial_2 + V_0 + x_3V_1, P]) &\leq \nu(P).\end{aligned}$$

Consider

$$S_n = (\text{ad}_{\tilde{L}_0})^n(\partial_1) \in \tilde{E}.$$

Their coefficients are polynomial in x_3 . Their real span is finite-dimensional, so a finite basis gives a common upper bound for $\nu(S_n)$.

Suppose $d \geq 3$, and write $\varepsilon(x_3) = ax_3^d + \dots$, where $a \neq 0$. Since $\deg r \leq d$, only $-ax_3^d\partial_3$ can increase weighted-degree by $d - 1$. Direct calculation gives

$$S_1 = k_1x_1\partial_3 + \text{terms of weighted-degree at most } 1$$

and iteratively,

$$[-ax_3^d\partial_3, x_1x_3^q\partial_3] = a(d - q)x_1x_3^{q+d-1}\partial_3.$$

Thus the highest-weighted-degree term of S_{n+1} is

$$k_1a^n \prod_{j=0}^{n-1} (d - j(d - 1))x_1x_3^{n(d-1)}\partial_3.$$

Every factor is nonzero because $d/(d - 1)$ is not an integer when $d \geq 3$. Its weighted-degree is $n(d - 1) + 2$, contradicting the common upper bound. Hence $d \leq 2$.

Step 4: Exclude $d \leq 2$ using the Wronskian lemma.

Let a be the coefficient of x_3^2 in ε . The preceding bounds give $\nu(S_n) \leq n + 1$. For $b(x_1, x_3)\partial_1^i\partial_3^j$ homogeneous of weighted-degree $n + 1$, direct calculation gives

$$\begin{aligned}[\tilde{L}_0, b\partial_1^i\partial_3^j] &= (\partial_1b)\partial_1^{i+1}\partial_3^j + (\partial_3b)\partial_1^i\partial_3^{j+1} \\ &\quad + ik_1x_1b\partial_1^{i-1}\partial_3^{j+1} + a(2jx_3b - x_3^2\partial_3b)\partial_1^i\partial_3^j \\ &\quad + \text{terms of weighted-degree at most } n + 1.\end{aligned}$$

The term multiplied by i is omitted when $i = 0$. The nonzero displayed terms have weighted-degree $n + 2$ and involve neither x_2 nor ∂_2 .

Starting from $S_0 = \partial_1$, the preceding commutator calculation shows inductively that the terms of weighted degree $n + 1$ in S_n have polynomial coefficients in x_1, x_3 and involve only ∂_1, ∂_3 . Write

$$S_n = \sum_{i,j \geq 0} b_{ij}^{(n)}(x_1, x_3) \partial_1^i \partial_3^j + \text{terms of weighted degree at most } n,$$

where the displayed sum consists of the terms of weighted degree $n + 1$. Introduce independent variables ξ, ζ and define

$$q_n(x_1, x_3, \xi, \zeta) = \sum_{i,j \geq 0} b_{ij}^{(n)}(x_1, x_3) \xi^i \zeta^j.$$

Thus $q_n = 0$ if and only if the weighted-degree- $(n + 1)$ terms of S_n vanish.

For a single term $b \partial_1^i \partial_3^j$, the terms of weighted-degree $n + 2$ in its commutator with $\tilde{L}_0$ are recorded by

$$(\partial_1 b) \xi^{i+1} \zeta^j + (\partial_3 b) \xi^i \zeta^{j+1} + i k_1 x_1 b \xi^{i-1} \zeta^{j+1} + a(2j x_3 b - x_3^2 \partial_3 b) \xi^i \zeta^j.$$

Here the term multiplied by i is omitted when $i = 0$. Summing over i, j gives

$$\begin{aligned} q_0 &= \xi, \\ q_{n+1} &= \xi \partial_{x_1} q_n + (\zeta - a x_3^2) \partial_{x_3} q_n + k_1 x_1 \zeta \partial_\xi q_n + 2a x_3 \zeta \partial_\zeta q_n. \end{aligned}$$

Terms of weighted degree at most n in S_n cannot contribute to this formula, since commutation with $\tilde{L}_0$ increases weighted degree by at most one.

To interpret this recurrence, consider the auxiliary system

$$\dot{x}_1 = \xi, \quad \dot{x}_3 = \zeta - a x_3^2, \quad \dot{\xi} = k_1 x_1 \zeta, \quad \dot{\zeta} = 2a x_3 \zeta.$$

Its right-hand sides are precisely the four coefficients in the recurrence. Along any local solution, the chain rule gives

$$\begin{aligned} \frac{d}{d\tau} q_n(x_1(\tau), x_3(\tau), \xi(\tau), \zeta(\tau)) &= \dot{x}_1 \partial_{x_1} q_n + \dot{x}_3 \partial_{x_3} q_n + \dot{\xi} \partial_\xi q_n + \dot{\zeta} \partial_\zeta q_n \\ &= q_{n+1}(x_1(\tau), x_3(\tau), \xi(\tau), \zeta(\tau)), \end{aligned}$$

where the partial derivatives are evaluated along the solution. Since $q_0 = \xi$, induction yields

$$q_n(x_1(\tau), x_3(\tau), \xi(\tau), \zeta(\tau)) = \frac{d^n \xi}{d\tau^n}(\tau).$$

Suppose now that $q_N = 0$ as a polynomial for some $N \geq 1$. Then $\xi^{(N)} = 0$ along every local solution. Consequently, $\xi(\tau)$ is polynomial in τ , and so is $x_1(\tau)$ because $\dot{x}_1 = \xi$.

The equations for x_3, ζ are independent of x_1, ξ . Fix $x_3(0) = 0$ and $\zeta(0) = 1$. By uniqueness, the two solutions with

$$(x_1(0), \xi(0)) = (1, 0) \quad \text{and} \quad (x_1(0), \xi(0)) = (0, 1)$$

have the same functions $x_3(\tau), \zeta(\tau)$. Their x_1 components therefore give polynomial solutions p, q of

the same equation

$$u'' = k_1 \zeta(\tau) u,$$

with $p(0) = 1, p'(0) = 0, q(0) = 0, q'(0) = 1$. Their Wronskian has derivative zero and initial value one, so $pq' - p'q = 1$ on a common interval, hence as a polynomial identity. Lemma 4.1 implies that their span contains the constant function 1. Substitution into the equation gives $k_1 \zeta(\tau) = 0$, contrary to $k_1 \neq 0$ and $\zeta(0) = 1$.

Therefore $q_n \neq 0$ for every n , and $\nu(S_n) = n + 1$, again contradicting the common upper bound. The assumed nonzero coefficient matrix is therefore impossible, and $k_1 = k_2 = h_2 = 0$ in the original coordinates. $\square$

Combining the proofs of

- (1) $b_1 = b_2 = 0$, i.e., ω_{12} is constant,
- (2) $k_3 = h_3 = 0$, i.e., ω_{13} and ω_{23} are independent with x_3 ,
- (3) $k_1 = k_2 = h_1 = h_2 = 0$, i.e., ω_{13} and ω_{23} are constants,

we then finish the establishment of the main theorem 1.1.

Appendix A. Proof of the Kernel of Euler Operator

Lemma A.1 (Global smooth Euler equation). *Let $Q \in \mathbb{C}[s] \setminus \{0\}$ and let $A = t\partial_t$. Then*

$$\ker_{C^\infty(\mathbb{R}; \mathbb{C})} Q(A) = \text{span}_{\mathbb{C}} \{t^n : n \in \mathbb{N}_0, Q(n) = 0\}.$$

In particular, every globally smooth solution of a nonzero polynomial Euler equation is a finite polynomial.

Proof. If Q is a nonzero constant, both sides of the asserted identity are zero. Assume henceforth that $N := \deg Q \geq 1$. For $u \in C^\infty(\mathbb{R}; \mathbb{C})$ and every $n \in \mathbb{N}_0$, the Leibniz rule gives

$$(Q(A)u)^{(n)}(0) = Q(n)u^{(n)}(0).$$

Suppose that $Q(A)u = 0$, and set

$$S = \{n \in \mathbb{N}_0 : Q(n) = 0\}, \quad p(t) = \sum_{n \in S} \frac{u^{(n)}(0)}{n!} t^n.$$

The set S is finite, $Q(A)p = 0$, and $v = u - p$ satisfies $Q(A)v = 0$ and is flat at $t = 0$.

We now rigorously exclude nonzero flat solutions. On $t > 0$, set $t = e^s$ and $V_+(s) = v(e^s)$; on $t < 0$, set $t = -e^s$ and $V_-(s) = v(-e^s)$. On both half-lines, $A = \partial_s$, and hence

$$Q(\partial_s)V_\pm = 0.$$

For either choice of sign, put

$$W_\pm(s) = (V_\pm(s), V'_\pm(s), \dots, V_\pm^{(N-1)}(s))^T.$$

After dividing the equation by the nonzero leading coefficient of Q , there is a constant complex companion matrix M such that $W'_\pm = MW_\pm$. Flatness of v and the chain rule imply

$$\|W_\pm(s)\| = o(e^{Ls}) \quad (s \rightarrow -\infty) \quad \text{for every } L > 0.$$

Choose a Euclidean norm on $\mathbb{C}^N$ and its induced matrix norm. Fix $s_0 \in \mathbb{R}$. Uniqueness for this constant-coefficient system gives, for $s \leq s_0$,

$$W_\pm(s_0) = e^{(s_0-s)M}W_\pm(s), \quad \|W_\pm(s_0)\| \leq e^{(s_0-s)\|M\|}\|W_\pm(s)\|.$$

Choose $L > \|M\|$ and let $s \rightarrow -\infty$. The right-hand side tends to zero, so $W_\pm(s_0) = 0$. Since s_0 was arbitrary, $V_+ = V_- = 0$. Consequently $v = 0$ on both half-lines and at the origin, proving the claim. $\square$

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